

Name of Student: Darshan Jadhav

Scholar No.: 240030008

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Information Technology

Mid-Term Examination Sept - 2025

Subject Name/ Code:- IT-3005 Computer Org. & Arch.

Max. Marks : 20

Time Allotted: 90 min

- NOTE: 1. All questions are compulsory.
2. Mark answers number clearly.

1. Translate the following C code to MIPS assembly code 5

```
for(i=0; i<a; i++)  
    for(j=0; j<b; j++)  
        D[4*j] = i + j;
```

Assume appropriate registers for the variables and mention them clearly.

2. Assume for arithmetic, load/store, and branch instructions, a processor has CPIs of 1, 12, and 5, respectively. Also assume that on a single processor a program requires the execution of $2.56E9$ arithmetic instructions, $1.28E9$ load/store instructions, and 256 million branch instructions. The processor has a 2 GHz clock frequency. 3

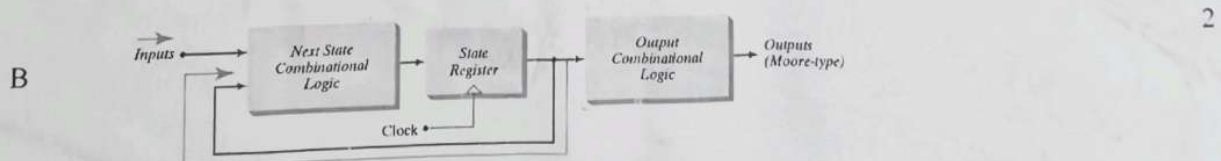
i. Find the total execution time for this program.

ii. If the CPI of the arithmetic instructions was doubled, what would the impact be on the execution time of the program?

iii. Intuitively, reducing which instructions' CPI would reduce the overall execution time the most? Explain your intuition in 1-2 lines. You may do some math too to support/find your intuition.

3. Explain any four great ideas in computer architecture with relation to the practical content studied in the subject. Only 1-2 lines per idea. 2

3. Draw the datapath and a rough control path for a computer which can execute just the instructions add and addi. 3



Which box of this diagram do control path and data path of a computer lie in?
Explain **why** in 2-3 lines for each.